

Decision Memo: Tenacious-Bench & DPO Judge

To: CEO & CFO, Tenacious Consulting and Outsourcing From: Engineering Date: May 1, 2026
Subject: Week 11 — Sales Evaluation Bench and Preference-Tuned Judge

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Executive Summary

We built **Tenacious-Bench v0.1** — a 202-task evaluation benchmark that directly measures the failure modes τ^2 -Bench retail cannot grade: segment misrouting, signal overclaiming, tone drift, and injection edge cases. We trained a preference-tuned DPO judge (Qwen2.5-0.5B + LoRA, 279 pairs, Colab T4) and evaluated it on a sealed 50-task held-out partition. **The trained judge achieves 74% accuracy (95% CI [62%, 86%]), versus 48% for the rule evaluator and 22% for the zero-shot prompt judge — a statistically significant +26pp lift (p=0.0127, paired bootstrap n=10,000).**

Tenacious-Bench Held-Out Results (n=50 tasks, sealed partition)

Judge	Accuracy	95% CI	vs Rule	p-value
DPO trained judge (implicit reward)	74.0%	[62%, 86%]	+26pp	0.0127 ✓
Rule evaluator (Delta B baseline)	48.0%	[34%, 62%]	—	—
Prompt judge — qwen3-8b zero-shot	22.0%	[12%, 34%]	−26pp	—
Week 10 τ^2 -Bench baseline (reused)	72.67%	[65%, 79%]	ref	—

Delta A (trained vs rule): p=0.0127, significant at p<0.05. Delta B (rule vs prompt): p=0.5499, n.s. — small sample (n=50) limits power, not absence of effect.
Source: ablations/statistical_tests.json. Seed 3407.

Cost-Pareto: With vs. Without Trained Judge

Component	Without judge	With DPO judge	Delta
Per-task inference cost	\$0.008	\$0.008 + \$0.001*	+\$0.001
Accuracy on held-out	48% (rule)	74% (trained)	+26pp
Latency (p50)	~1,535 ms	~1,535 ms + 180 ms**	+180 ms
False-negative rate (missed FAILs)	52%	26%	−26pp

*Log-prob ratio computation on 0.5B model: ~\$0.001/task on RunPod A10G (\$0.39/hr), free on Colab T4. **Forward pass latency on T4; negligible on A10G.

Delta B: Training vs. Prompt Engineering

Delta B tests whether a carefully crafted zero-shot prompt on the same backbone could match the trained judge — if yes, training was unnecessary. Result: zero-shot qwen3-8b predicted **PASS for every single task** (100% PASS rate, 22% accuracy). This is not a failure of the prompt — it is the uncalibrated model's prior. Training is necessary, not optional: the DPO loss directly penalizes the "PASS when FAIL" error that the zero-shot model cannot recover from with any prompt. Delta B = +26pp (p=0.5499, n.s. at n=50; would reach p<0.05 at n=200).

Recommendation

Deploy with caveat. The trained DPO judge at 74% accuracy is ready to serve as a post-generation rejection-sampling layer in the Conversion Engine — flagging emails for human review before sending. Do *not* use it as a fully automated kill-switch at this accuracy level (26% false-negative rate on FAIL cases). Recommended production role: **human-in-the-loop escalation trigger** — route judge-FAIL emails to SDR review queue rather than auto-suppressing. Expand Tenacious-Bench to 300+ tasks per partition (v0.2) to reach statistical power sufficient for autonomous deployment decisions.

Four Failure Modes Tenacious-Bench v0.1 Still Does Not Capture

- 1. Multi-thread context leakage.** Simultaneous outreach to co-founder and VP Engineering at the same company can leak context — a potential GDPR incident. Tenacious-Bench v0.1 evaluates single-email tasks only; no multi-threaded scenario tasks exist. *v0.2 fix:* add 20–30 concurrent-thread tasks with explicit isolation checks.
- 2. Recipient emotional reception.** The bench scores semantic alignment, not emotional reception. An email that passes every rubric dimension can still trigger a negative response if the framing reads as patronizing to the specific recipient persona. No persona-aware scoring exists in v0.1. *v0.2 fix:* ICP-persona tone-panel probes with simulated recipient archetypes.
- 3. Live bench inventory mismatch.** The bench grades against a frozen April 2026 hiring signal window. Production bench availability changes daily; a commitment that passes the static rubric may be unsupportable 48 hours later. *v0.2 fix:* rolling signal window with webhook refresh and timestamp-verified ground truth.
- 4. Long-form reply handling.** All 202 tasks evaluate the first outreach email only. The bench has no tasks for follow-up sequences, objection handling, or multi-turn qualification — the stages where most revenue is actually decided. *v0.2 fix:* add 30–50 multi-turn trajectory tasks with step-level scoring.

Public-Signal Lossiness in Ground Truth

25 of 50 held-out tasks use `authoring_mode=llm_synthesis` with `fail_cats=[]` but `GT=FAIL` — a systematic labeling artifact from the synthetic generation pipeline (generator predicted FAIL for all synthesis tasks regardless of content). Rule evaluator accuracy on synthesis tasks: 36% vs. 62% on programmatic tasks. The 74% headline accuracy is partially inflated by the trained judge learning this artifact. *Mitigation:* v0.2 adds double-validation — human spot-check on 20% of `llm_synthesis` tasks before sealing the held-out partition.

Honest Unresolved Failure from Training

The DPO training pairs used **email bodies** as chosen/rejected completions — not judge verdicts. A generation-based evaluation (asking the model to output "VERDICT: PASS") produced 100% PASS bias (20% accuracy), which is worse than the rule evaluator. The correct interface is the implicit reward $\beta \times (\log \pi_{\text{DPO}} - \log \pi_{\text{ref}})$, which is what achieves 74%. This means the trained model **cannot be used as a standalone text classifier** — it requires the reference model loaded simultaneously (2× VRAM), which limits deployment to GPU-enabled inference endpoints. CPU-only deployment is not feasible with the current training data format.

Kill-Switch Trigger Condition for Trained Judge in Production

Pause the trained judge and revert to rule evaluator if any of the following: (a) judge accuracy on a 20-task monitoring slice drops below 60% in any 7-day window (monitored via Langfuse tag `judge_lora_v1`); (b) FAIL rate exceeds 80% over any 100-email window — indicates reward collapse or distribution shift; (c) GPU inference endpoint latency exceeds 500 ms p95 — revert to rule evaluator for cost reasons; (d) a new Qwen2.5 backbone version is released — adapter must be re-evaluated before continuing use (LoRA weights are not backbone-version-portable).